

Andrew Erskine

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Summary

- Systems neuroscientist with over 10 years experience in in-vivo extracellular electrophysiology, 2p imaging and animal behaviour.
- Proficient software engineer (.NET, Python, MATLAB, bonsai-rx) with focus on building neuroscience research infrastructure: experiment monitoring; data management; hardware integration; analysis pipelines.
- Highly skilled in surgical techniques for chronic and acute mammalian neurophysiology: electrode implantation; chronic window implants; head-fixation; viral injection.
- Dedication to writing clean, production-quality code and comfortable with unit testing, efficient version control, CI/CD and DevOps practices.
- Experienced in open-source software contributions, including production and dissemination of documentation.
- Extensive training in computational/analysis approaches for neural and behavioural data (spike-sorting, population analyses / dimensionality reduction, cell segmentation, behaviour tracking)

Experience

NeuroGEARS Ltd.

London, United Kingdom

SENIOR SOFTWARE ENGINEER

2022 - Present

Software / consulting company developing the bonsai-rx language and custom software tools for neuroscience research.

- Contributor to the popular open source bonsai-rx language for visual reactive programming.
Built and released Bonsai libraries for hardware integration (Aerotech, NewScale, OpenEphys/ONIX), networking (ZeroMQ, Zyre, Lsl) and image acquisition (EmergentTech).
- Neuroscience/software consultant.
Produced experimental solutions for clients in top research labs (Sainsbury Wellcome Centre, Allen Institute, University College London, Cambridge University) including hardware/software infrastructure; code and data standardisation; experiment design; hardware integration and synchronisation (OpenEphys, NeuroPixels, National Instruments, Zaber, Harp)
- Aided in hardware/firmware design of custom Harp protocol devices for sophisticated closed-loop experimental rigs with millisecond-level round-trip latency requirements.

🔧 `bonsai-rx` · `C#` & `.NET` · `Python` · `MATLAB` · `Unity` · `Godot` · `ZeroMQ` · `Avalonia` · `Windows Forms` · `MAUI` · `Git`

The University of Southern California

Los Angeles, California

POSTDOCTORAL SCHOLAR

2018 - 2022

Investigating somatosensory processing in neural circuits with 2p imaging and 3D optogenetics.

- Employed in-vivo all-optical techniques (2p, SLM optogenetic stimulation) to investigate neuronal ensemble recruitment in somatosensory cortex.
- Surgical preparations for chronic imaging during head-fixed behavior over several weeks (window implant, viral injection, head-fixation).
- Reduced manual analysis time by modifying DeepLabCut for Google Cloud, allowing for fast, parallel usage on TB size whisker tracking datasets and speedup of data processing.
- Applied deep neural network models with dimensionality reduction methods to analyze neural population responses in high-dimensional space.
- Designed and deployed machine-learning pipelines to increase analysis throughput in the lab (Google Cloud, Colab).
- Mentored graduate and undergraduate students and provided training in data analysis and programming.

🔧 `Python` · `MATLAB` · `Keras` · `tensorflow` · `numpy` · `pandas` · `jupyter`

The Francis Crick Institute / University College London

London, United Kingdom

PHD STUDENT

2013 - 2018

Building automated systems for mouse behavioral studies and investigating the temporal component of olfaction.

- Redesigned high throughput mouse behavior system (AutonoMouse) in Python – including sensor interfaces, experiment control and database.
- Surgical preparations for in-vivo multi-channel silicon probe electrophysiology.
- Developed several auxiliary Python libraries that became standard tools: `daqface` for communicating with National Instruments ADCs and `PulseBoy` for designing complex digital command patterns.
- Designed a novel odor-delivery device and software package for flexibly generating complex valve patterns with modular design (`PulseBoy`).

🔧 `Python` · `MATLAB` · `Qt` · `nidaqmx`

University of Manchester

Manchester, United Kingdom

MASTERS STUDENT

2012 - 2013

Developed an experimental rig for extracellular recording (tungsten microelectrodes) of trigeminal ganglion neurons and high-speed whisker kinematic tracking during active, awake behaviour.

Education

University College London

London, United Kingdom

PHD IN NEUROSCIENCE

2018

- Thesis: Perception and representation of temporally patterned odor stimuli in the mammalian olfactory bulb

University of Manchester

Manchester, United Kingdom

MNEUROSCI, FIRST CLASS HONOURS

2013

- Thesis: Representation of whisker kinematic parameters in the trigeminal ganglion of awake, behaving mice

Courses

Cold Spring Harbor Laboratory

Cold Spring Harbor, New York

IMAGING STRUCTURE AND FUNCTION IN THE NERVOUS SYSTEM

2019

- Intensive training in advanced optical methods including spatial-light modulation, temporal focusing, light-sheet, acousto-optical deflectors, remote-focusing, SLAP, SCAPE.

SysMIC

London, United Kingdom

SysMIC COURSE

2015

- Advanced training in practical applications of mathematics and computing in biology. Final project: MATLAB implementation of FitzHugh-Nagumo model of single neuron activity.

Bernstein Center Freiburg

Freiburg, Germany

ANALYSIS AND MODELS IN NEUROPHYSIOLOGY

2014

- Instruction in computational neuroscience theory and application relating to electrophysiological data.

Publications

- Aeon: an open-source platform to study the neural basis of ethological behaviours over naturalistic timescales.**, D. Campagner, Bhagat, J., Lopes, G., Calcaterra, L., Pouget, A. G., Almeida, A., Nguyen, T. T., Lo, C. H., Ryan, T., Cruz, B., Carvalho, F. J., Li, Z., Erskine, A., Rapela, J., Folsz, O., Marin, M., Ahn, J., Nierwetberg, S., Lenzi, S. C., Reggiani, J. D. S., SGEN group – SWC GCNU Experimental Neuroethology Group. *bioRxiv*
- Fast odour dynamics are encoded in the olfactory system and guide behaviour.**, Ackels, T., Erskine, A., Dasgupta, D., Marin, A. C., Warner, T. P. A., Tootoonian, S., Fukunaga, I., Harris, J. J., Schaefer, A. T. *Nature*
- Behavioral and neural bases of tactile shape discrimination learning in head-fixed mice.**, Kim, J., Erskine, A., Cheung, J. A., Hires, S. A. *Neuron*
- AutonoMouse: High throughput operant conditioning reveals progressive impairment with graded olfactory bulb lesions.**, Erskine, A., Bus. T., Herb, J. T., Schaefer, A. T. *PLOS ONE*
- Prediction of primary somatosensory neuron activity during active tactile exploration.**, Campagner, D., Evans, M. H., Bale, M. R., Erskine, A., Petersen, R. S. *eLife*
- Microsecond-Scale Timing Precision in Rodent Trigeminal Primary Afferents.**, Bale, M. R., Campagner, D., Erskine, A., Petersen, R. S. *JNeurosci*

Writing

A practical guide to RNNs for neuroscience research in Keras

2021

Article covering practical implementations and applications of recurrent neural network models for neuroscience research.

Unity multiplayer: bottom to top

2021

Tutorial outlining the development of a full-stack multiplayer Unity app. Now featured as part of the official documentation for the DarkRift multiplayer framework.